

Stream Control Transmission Protocol (SCTP) - Detailed Notes

Overview

SCTP is a transport layer protocol like TCP and UDP. It is message-oriented (preserves message boundaries) and reliable. SCTP is designed to overcome TCP and UDP limitations such as head-of-line blocking in TCP and lack of reliability in UDP.

Performance Comparison with UDP and TCP

Protocol	Performance Characteristics	Best Use Cases
UDP	Low delay, no connection setup, no retransmission	Real-time apps (VoIP, video calls, gaming)
TCP	Reliable, flow control and congestion control, may suffer from head-of-line blocking	File transfers, email, web browsing
SCTP	Reliable, faster recovery than TCP, multi-streaming	Signaling, multimedia, telephony

SCTP Services

1. Process-to-process communication: Like TCP, reliable communication between processes.
2. Multiple streams: Independent data flows within one connection, preventing head-of-line blocking.
3. Multihoming: A host can have multiple IP addresses for fault tolerance.
4. Full-duplex communication: Data flows both ways simultaneously.
5. Connection-oriented: Uses handshake like TCP.
6. Reliable service: Uses sequence numbers and acknowledgments.

SCTP Features

Feature	Description
Transmission Sequence Number (TSN)	Ensures reliable delivery of all chunks.
Stream Identifier (SID)	Identifies multiple streams in a connection.
Stream Sequence Number (SSN)	Maintains order within each stream.
Packets	Contain one or more data chunks.
Acknowledgment Number	Confirms received data.
Flow Control	Manages data rate between sender and receiver.
Error Control	Detects and retransmits lost or corrupted data.
Congestion Control	Similar to TCP (slow start, congestion avoidance).